

CSC/SDS 109: Communicating with Data

a3: Hand-drawn Visualization—CEEDS Survey Data

Goals:

- Find interesting trends or patterns in the CEEDS Sustainability Survey Data
- Compose a set of hand-drawn visualizations illustrating your findings

Important Notes:

- Notice that part 1 of this assignment spans 2 weeks. The quality of what you produce should reflect two full weeks of work; do your best to deliver visualization(s) that are creative, precise, and meaningful.
- You may work alone or in small groups (2-3). If you choose to work in a group, the quantity and quality of your visualization(s) should scale accordingly.

Instructions – Part 1 DUE IN CLASS ON VIEWING DAY

For this assignment, you will analyze data from CEEDS: Smith_Sustainability_2024_Sep.xlsx. You can find a link to the dataset on the course website under “Assignments”. CEEDS collected this data through a survey of Smith students with the goal of understanding attitudes and perceptions of sustainability across campus.

Familiarize yourself with the data. What questions were asked in the survey? What options were available for answering each question?

Hypothesize which variables might reveal interesting patterns or have relationships to each other. Investigate your hypotheses using R (we started code in class), Excel, or any program you are comfortable with. You do not need to do formal statistical testing, looking at distributions etc. is sufficient for this assignment.

Create three visualizations, by hand, to illustrate your findings. Each visualization must be unique, no more than 1 variable can be shared between visualizations, and each must have a distinct visual encoding. Two of the three visualizations must encode at least three variables.

Competency Demonstration – Part 2, start in class on viewing day

On viewing day (see course calendar), bring your complete visualization(s) to class. Viewing day will be reserved to engage with each other's visualizations and provide feedback. You will also get verbal feedback from the instructor on viewing day. One way to provide feedback and demonstrate competencies is to write a thoughtful and actionable critique of someone else's visualization(s). If you choose to do this, be sure to cite visualization science and theories to support your suggested changes and compliments.

This is a competency-based course. After you complete your assignment, write a short, succinct, and clear reflection explaining which core competencies you demonstrated in your work and how.

It is recommended that you use this assignment to demonstrate 2-3 of the following competencies:

- Map data to appropriate visual channels
- Apply perceptual concepts to visualization design
- Accurately encode data values via color
- Write a thoughtful and actionable critique of someone else's work
- Redesign a visualization based on critiques
- Positively contribute to the classroom community

If you choose to demonstrate "Write a thoughtful and actionable critique of someone else's work", include your critique and an image of the visualization it critiques in your submission.

If you choose to demonstrate "Redesign a visualization based on critiques", include the critiques you received and a clear, concise, detailed sketch/outline of how you would revise your visualization based on the critiques.

Submission

Create a PDF with images of your visualization(s) and your write up from part 2. Submit your completed work as a PDF on Gradescope.